MILITARY SPECIFICATION SHEET

RESISTORS, VARIABLE, WIRE-WOUND

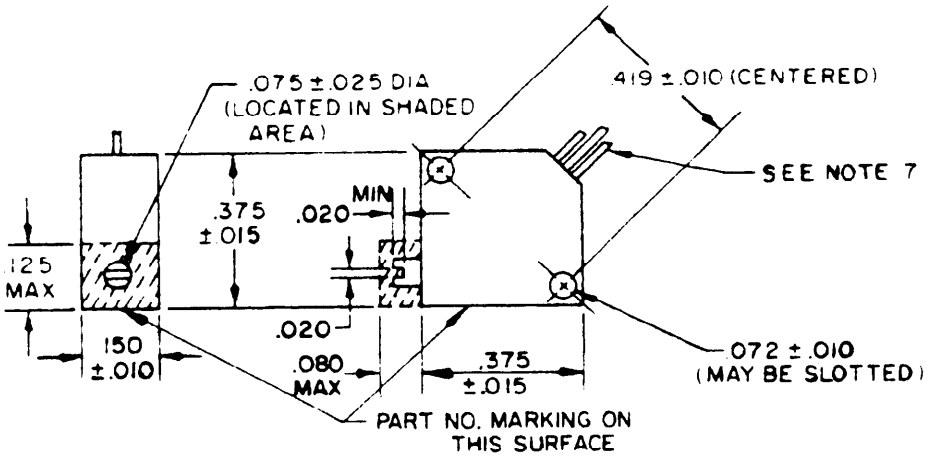
(LEAD SCREW ACTUATED),

ESTABLISHED RELIABILITY

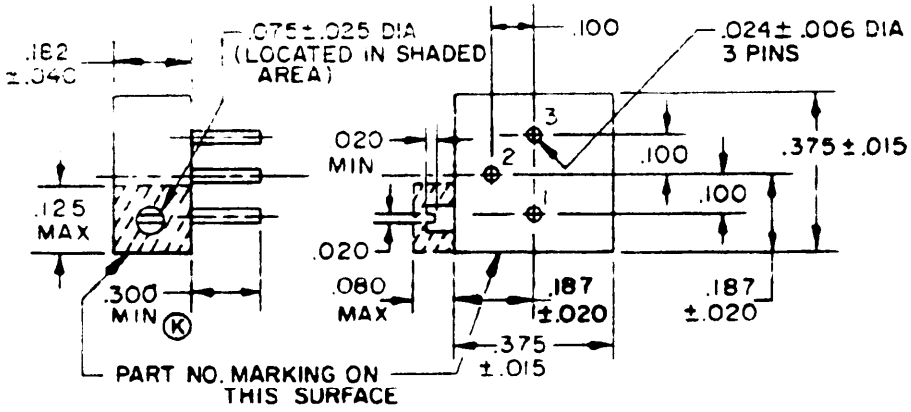
STYLE RTR24

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of Specification MIL-R-39015.

This specification is approved for use by all Departments and Agencies of the Department of Defense.



TERMINAL TYPE L



TERMINAL TYPE P

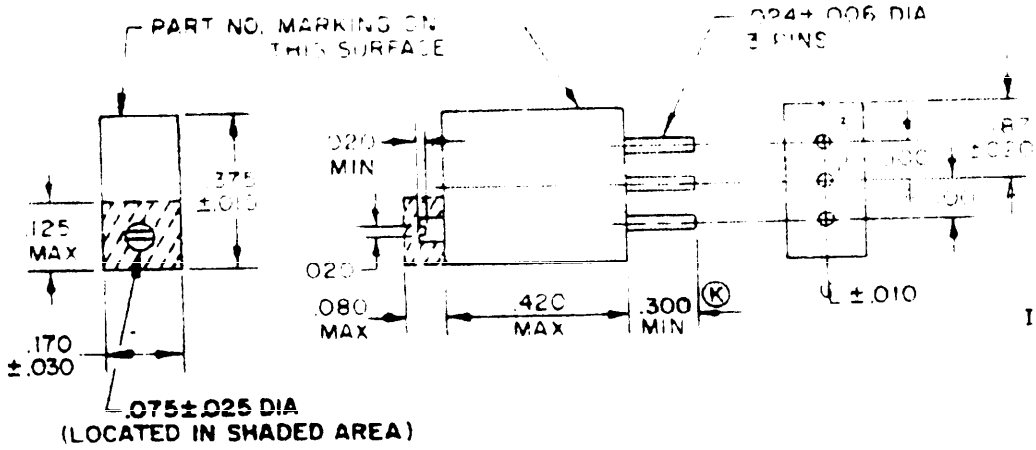
INCHES	MM
.006	.15
.010	.25
.015	.38
.020	.51
.024	.61
.025	.64
.040	1.02
.072	1.83
.075	1.91
.080	2.03
.100	2.54
.125	3.18
.150	3.81
.182	4.62
.187	4.75
.300	7.62
.375	9.53
.419	10.64

FIGURE 1. Style RTR24 resistors.

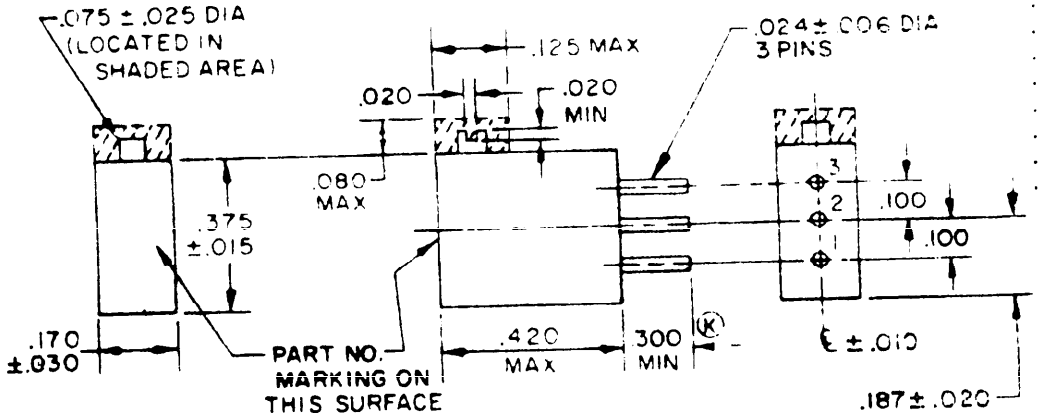
(K) denotes changes

FSC 5905

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TERMINAL TYPE W



TERMINAL TYPE X

INCHES	MM
.006	.15
.010	.25
.015	.38
.020	.51
.024	.61
.025	.64
.030	.76
.075	1.91
.080	2.03
.100	2.54
.125	3.18
.170	4.32
.187	4.75
.300	7.62
.375	9.53
.420	10.67

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005(.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The entire slot of the actuating screw must be above the surface of the unit.
5. For types P, W, and X, normal mounting means is by use of pins only.
6. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
7. The three leads shall be stranded wire, AWG size 28 to 30, having minimum length of 6(152.4 mm); they shall be insulated with polytetrafluoroethylene stripped .250±.062(6.35±1.57 mm) from the end, and color coded.
8. Maximum weight is .00286 pound(1.3 grams).

FIGURE 1. Style RTR24 resistors - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure.

Resistance values: See table.

Power rating: 3/4 watt at full load operation at an ambient temperature of 85°C.

Maximum resolution: See table.

Rated working voltage: See table.

Operating torque:
Maximum - 5 ounce-inches.

Actual effective-electrical travel:
15 turns minimum.
30 turns maximum.

Conditioning: Method 108 of MIL-STD-202. ΔR = 0.5 percent for resistance values of 100 ohms or greater, and ± (0.5 percent + 0.05 ohm) for values below 100 ohms.

Peak noise: Not to exceed 500 ohms.

DC resistance: Method 303 of MIL-STD-202.

Resistance tolerance: ±5 percent.

Dielectric withstanding voltage:
At atmospheric pressure: Method 301 of MIL-STD-202. Test voltage at 900 volts.
At reduced barometric pressure:
Method 105, test condition C, MIL-STD-202.
Test voltage: 350 volts.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 1,000 megohms.

Thermal shock: Method 107, test condition B, MIL-STD-202. ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Solderability (applicable to terminal P, W, and X types only): Method 208 of MIL-STD-202.

Resistance temperature characteristic:
Method 304 of MIL-STD-202. ±50 PPM °C.

Moisture resistance: Method 106 of MIL-STD-202. ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Shock (specified pulse): Method 213 of MIL-STD-202. ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Vibration, high frequency: Method 204 of MIL-STD-202. ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Salt spray (corrosion): Method 101, test condition A, MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202. ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Low temperature operation: ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Low temperature storage: ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

High temperature exposure: ΔR = 1 percent for resistance values of 100 ohms or greater, and ± (1 percent + 0.05 ohm) for values below 100 ohms.

Rotational life: ΔR = ±2 percent.

Life: Method 108 of MIL-STD-202. ΔR = ±2, ±3 percent.

Resistance to solvents: Method 215 of MIL-STD-202.

Terminals: Flexible lead type terminals shall conform to MIL-W-16878.

Minimum resistance value: 10 ohms.

Maximum resistance value: 10,000 ohms.

Terminal types:
Printed circuit types - P, W, and X.
Flexible lead type - L.

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Failure rate level designation
In percent/1,000 hours.

M = 1.0

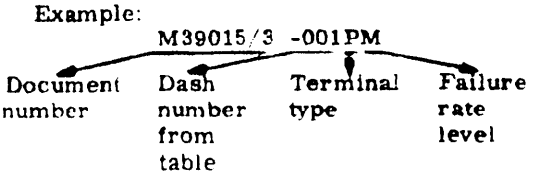
P = 0.1

R = 0.01

S = 0.001

Part number: Consists of the number of this specification sheet and a dash number with letters which signify. Terminal and failure rate level.

Example:



Style RTR24

Type designation 1/	Nominal resistance value	Maximum resolution 2/	Maximum rated ac and dc working voltage	Dash number 3/
	Ohms	Percent	Volts	
RTR24D-100-	10	1.3	2.7	001
RTR24D-200-	20	1.1	3.8	002
RTR24D-500-	50	.77	6.1	003
RTR24D-101-	100	.62	8.7	004
RTR24D-201-	200	.55	12.3	005
RTR24D-501-	500	.51	19.4	006
RTR24D-102-	1,000	.37	27.4	007
RTR24D-202-	2,000	.30	38.7	008
RTR24D-502-	5,000 4/	.25	61.3	009
RTR24D-103-	10,000 5/	.19	86.7	010

1/ Complete type designation is for information only and will include additional symbols to complete type designation.
2/ Maximum resolutions shown are theoretical.
3/ Complete dash number will include symbols for termina type and failure rate level.
4/ Value based on the use of wire having no less than 0.001-inch ±10 percent diameter.
5/ Value based on the use of wire having no less than 0.0008-inch ±10 percent diameter.

(K) Style RTR24 resistors with terminal types P, W, and X are interchangeable with resistors having the same terminal designation under MIL-R-39015/3J. The changes in dimensions do not require the assignment of a new FSN where a Federal Stock Number had previously been assigned to these items under Revisions E, F, G, H, or J of this specification.

MIL-R-27208 substitution data: Resistors of this specification regardless of their failure rate designation are substitutes for resistors of the same resistance value, tolerance, terminal characteristic, and resistance temperature characteristic specified in the inactivated specification MIL-R-27208/9.

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Navy - EC
Air Force - 11

Review activities:
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Navy - EC, AS, OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy -
Air Force - 19

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5905-0836)

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